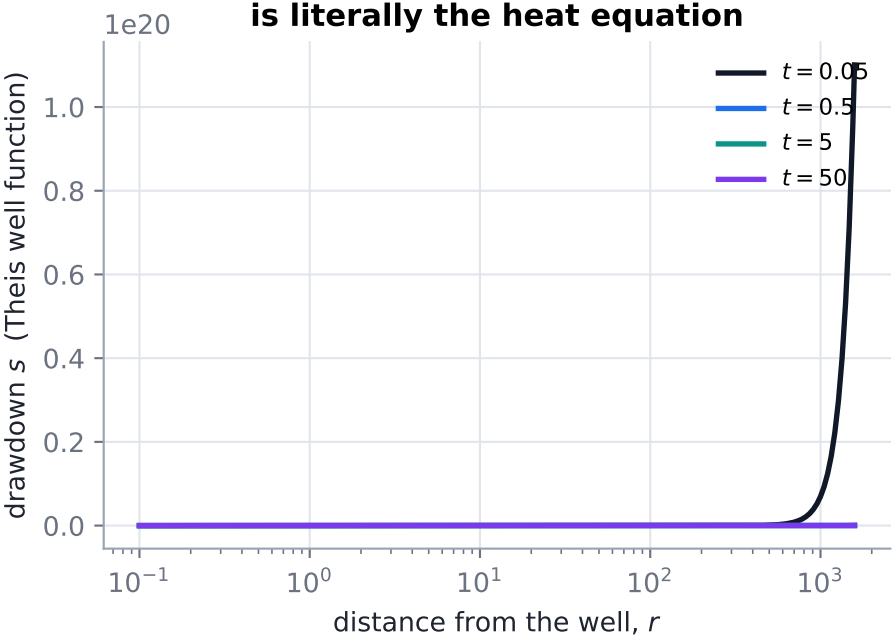
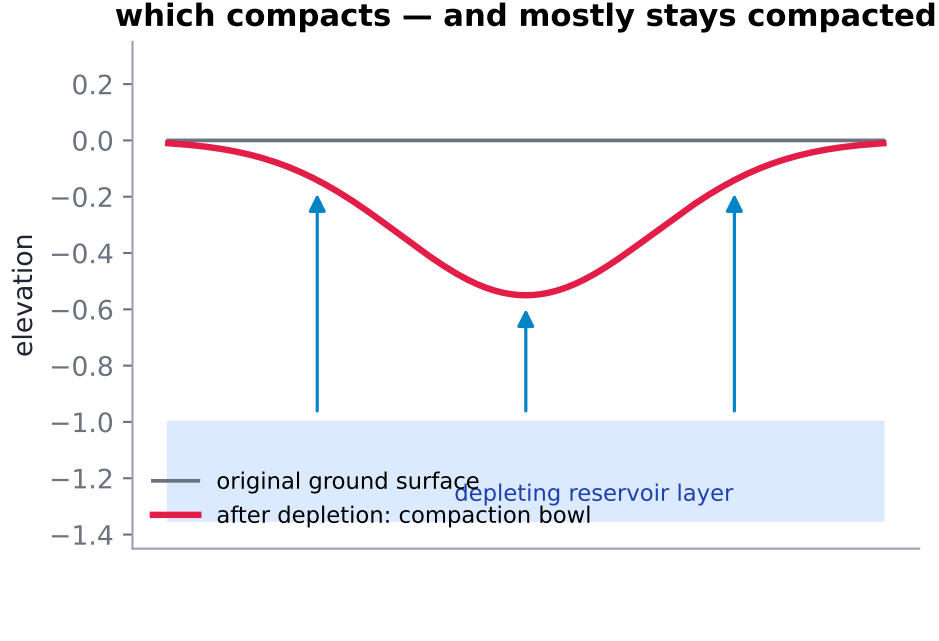

pressure diffuses outward
the aquifer equation $\partial_t s = (T/S)\nabla^2 s$
is literally the heat equation
1e20
drawdown s (Theis well function)
distance from the well, r
$t = 0.05$
$t = 0.5$
$t = 5$
$t = 50$


effective stress does the rest
$\sigma' = \sigma - p$: lower pressure loads the matrix,
which compacts — and mostly stays compacted
elevation
original ground surface
after depletion: compaction bowl
depleting reservoir layer